

Wenjing Gong

Ph.D. student
Urban and Regional Science
College of Architecture
Texas A&M University

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(Updated on 05/2026)

RESEARCH INTERESTS

- Urban Analytics, Urban Planning, GIScience, GeoAI
- Climate Resilience, Environmental Health, Urban Sustainability

EDUCATION

- 2024 – pres. Ph.D. Student, Urban and Regional Science (Urban Analytics & GeoAI)**
Texas A&M University, College Station, USA
Chair: Dr. Lingqian (Ivy) Hu, Co-Chair: Dr. Xinyue Ye,
Committee members: Dr. Galen Newman and Dr. Lei Zou
- 2026 – pres. Graduate Certificate in Geographic Information Science**
Department of Geography, Texas A&M University, College Station, USA
- 2019 – 2022 M.E., Architecture (Urban Study)**
Tongji University, Shanghai, China
GPA: 88.5/100 (Entrance through the exam-free recommendation program)
- 2019 Visiting Student, Graduate School of Architecture, Planning and Preservation**
Columbia University, New York, USA
- 2014 – 2019 B.E., Architecture**
Shandong University of Science & Technology, Qingdao, China
GPA: 90.5/100 (Rank: 1/60 each year)

ACADEMIC & PROFESSIONAL EXPERIENCE

- 2025 – 2026 Student Ambassador**
Texas A&M Institute of Data Science, College Station, USA
- 2025 – 2026 Student Fellow**
Telehealth Institute, Texas A&M University, College Station, USA
- 2024 – 2025 Research Assistant**
Advisor: Dr. Xinyue Ye, Center for Geospatial Sciences, Applications and Technology,
Texas A&M University, College Station, USA
- 2022 – 2024 Designer and Planner**
Architecture Design & Research Institute of Tongji University, Shanghai, China

PEER-REVIEWED PUBLICATIONS

* Denotes corresponding author.

- 2026 Rui, J., Shabrina, Z., & **Gong, W.*** (2026). Artificial intelligence applications in urban extreme heat management: A systematic review of forecasting, monitoring, mitigation and decision support. *Environmental Impact Assessment Review*, 119, 108363. <https://doi.org/10.1016/j.eiar.2026.108363>
- 2026 Ye, X., **Gong, W.**, Yang, Y., Zou, L., Tu, Z., Huang, X., ... & Wu, L. (2026). Towards Agentic Urban Digital Twins (AUDiTs): advancing new urban science through Human-AI co-learning agents. *Urban Informatics*, 5(1), 9. <https://doi.org/10.1007/s44212-025-00099-3>
- 2026 Rui, J., & **Gong, W.*** (2026). Social media-driven multi-scale flood sensing: A systematic review of applications from global monitoring to community resilience. *Journal of Environmental Management*, 407, 129895. <https://doi.org/10.1016/j.jenvman.2026.129895>
- 2026 Reed, K., Rybkowski, Z., Bardenhagen, E., **Gong, W.**, Starrett, C., Ennemoser, B., & Ye, X. (2026). Enhancing interdisciplinary competence in built environment research: Evaluating the impact of a two-day design charrette on undergraduates. *Frontiers of Urban and Rural Planning*, 4(1), 2. <https://doi.org/10.1007/s44243-025-00073-2>
- 2026 **Gong, W.**, Wu, L., Zhu, C., Song, Y., & Ye, X. (2026). Revealing park visitation under dual environmental threats in a socially stratified city: Evidence from smartphone mobility data in Dallas. *Urban Forestry & Urban Greening*, 117, 129264. <https://doi.org/10.1016/j.ufug.2026.129264>
- 2026 Rui, J., & **Gong, W.** (2026). Paying lip service? An investigation into the spatial mismatch between younger and older adults' streetscape perceptual preference and visitation behavior. *Cities*, 171, 106750. <https://doi.org/10.1016/j.cities.2025.106750>
- 2025 **Gong, W.**, Ye, X., Wu, K., Jamonnak, S., Zhang, W., Yang, Y., & Huang, X. (2025). Integrating Spatiotemporal Vision Transformer into Digital Twins for High-Resolution Heat Stress Forecasting in Campus Environments. *Journal of Planning Education and Research*, 0739456X251391121. <https://doi.org/10.1177/0739456X251391121>
- 2025 Ye, X., Yigitcanlar, T., Xu, Y., Head, M., Hang, Y., Sanchez, T., **Gong, W.**, & Niyogi, D. (2025). Urban planning and IPCC-like city assessments integration for climate-resilient cities. *Environment and Planning B*, 23998083251330940. <https://doi.org/10.1177/23998083251330940>
- 2025 Ye, X., Yigitcanlar, T., Goodchild, M., Huang, X., Li, W., Shaw, S.-L., Fu, Y., **Gong, W.**, & Newman, G. (2025). Artificial intelligence in urban science: Why does it matter? *Annals of GIS*, 0(0), 1–9. <https://doi.org/10.1080/19475683.2025.2469110>
- 2025 Ye, X., Li, S., **Gong, W.***, Li, X., Li, X., Dadashova, B., Li, W., Du, J., & Wu, D. (2025). Street View Imagery in Traffic Crash and Road Safety Analysis: A Review. *Applied Spatial Analysis and Policy*, 18(2), 50. <https://doi.org/10.1007/s12061-025-09653-7>
- 2025 **Gong, W.**, Lee, C. S., Li, S., Adkison, D., Li, N., Wu, L., & Ye, X. (2025). Cyber victimization in hybrid space: An analysis of employment scams using natural language processing and machine learning models. *Journal of Crime and Justice*, 1–22. <https://doi.org/10.1080/0735648X.2024.2448804>
- 2024 **Gong, W.**, Rui, J., & Li, T. (2024). Deciphering urban bike-sharing patterns: An in-depth analysis of natural environment and visual quality in New York's Citi bike system. *Journal of Transport Geography*, 115, 103799. <https://doi.org/10.1016/j.jtrangeo.2024.103799>
- 2023 **Gong, W.**, Huang, X., White, M., & Langenheim, N. (2023). Walkability Perceptions and Gender

Differences in Urban Fringe New Towns: A Case Study of Shanghai. *Land*, 12(7), 1339.
<https://doi.org/10.3390/land12071339>

PRESENTATIONS

Conferences

- 2026 **Gong, W.** Toward Climate-Responsive Campuses: A Digital Twin Framework for High-Resolution Heat Stress Forecasting and Planning. *2026 Association of Collegiate Schools of Planning*, October 8-10, 2026, Pittsburgh, Pennsylvania, USA (Oral).
- 2026 **Gong, W.** Hyperlocal PM_{2.5} Imputation from Sparse Sensors with Earth Embeddings and State Space-Enhanced Spatial Learning. *2026 International Symposium on Spatiotemporal Data Science*, August 7-8, 2026, Virginia Tech, Virginia, USA (Oral).
- 2026 **Gong, W.**, Ye, X., Song, Y. Essential Infrastructure or Unavoidable Risk? Modeling Urban Park Visitation under Dual Climate and Crime Stressors with Large-Scale Mobility Data. *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA (Oral).
- 2025 **Gong, W.**, Ye, X., Wu, K., Jamonnak, S., Zhang, W., Yang, Y., Huang, X. Integrating Spatiotemporal Vision Transformer into Digital Twins for High-Resolution Heat Stress Forecasting in Campus Environments. *2025 Annual Meeting of American Association of Geographers*, March 24-28, 2025, Detroit, Michigan, USA (Oral).
- 2024 **Gong, W.**, Yang, Y., Zhang, W., Wu, K., Ye, X., Huang, X. Integrating high-resolution simulations and transformer network predictions for mitigating human heat stress: A digital twin of a Texas campus. *2024 Texas Weather Conference*, November 9, 2024, College Station, Texas, USA (Oral).
- 2024 **Gong, W.**, Yang, Y., Zhang, W., Wu, K., Ye, X., Huang, X. Integrating high-resolution simulations and transformer network predictions for mitigating human heat stress: A digital twin of a Texas campus. *College of Architecture's 26th Annual Showcase*, October 3, 2024, Texas A&M University, College Station, Texas, USA (Poster).
- 2024 **Gong, W.** Simulating and Predicting Spatial-Temporal Human Outdoor Heat Exposure: A Case Study on a Texas Campus. *The 2024 Symposium on Spatiotemporal Data Science: GeoAI for Social Sciences*, July 24, 2024, Virtual (Abstract accepted).

Workshops

- 2025 Yang, Y., **Gong, W.**, Zhang, K. DisasterVLP: Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models. *Summer School on Cyberinfrastructure and Disaster Resilience*, August 15, 2025, Texas A&M University, College Station, USA (Oral).
- 2022 Gu, J., **Gong, W.**, Zhang, X. Analysis of the Spatial Distribution of Express Courier Stations and Assessment of Site Selection Effectiveness. *DigitalFUTURES 12th Summer Workshops*, July 2, 2022, Virtual (Oral).
- 2019 **Gong, W.**, Huang, H., Zhang, X. Undoing Sprawl: Urban Agriculture as a Social Innovation. *Resizing the Urban Form in the Era of "Negative Population Growth" Workshop*, December 9, 2019, Columbia University, New York City, USA (Oral).

AWARDS & HONORS

2026	Environmental Perception and Behavior Geography (EPBG) Travel Grant, 2026 Annual Meeting of American Association of Geographers, \$150.
2026	International Geographic Information Fund (IGIF) Scholarship Award, 2026 Annual Meeting of American Association of Geographers, \$1,200.
2026	Applied Geography Specialty Group Annual Meeting Award, 2026 Annual Meeting of American Association of Geographers, \$202.
2026	Travel Grant, LAUP Department, TAMU, \$600.
2026	2025–2026 NSF I-GUIDE Spatial AI Challenge — GeoAgent4Disaster (Accepted Project).
2025	Travel Grant, Summer School on Cyberinfrastructure and Disaster Resilience, TAMU (funded by NSF), \$200.
2025	Texas A&M Institute of Data Science (TAMIDS) Student Ambassador Scholarship, Domain Data Science Track (2025–2026 cohort), TAMU, \$2,000.
2025	Ashok K. Dutt Award for Best Graduate Student Paper (1st place), Regional Development and Planning Specialty Group, 2025 Annual Meeting of American Association of Geographers, \$200.
2025	UCGIS I-GUIDE 2024–2025 Community Champions, \$3,000 (Student leader; project under Dr. Xinyue Ye).
2025	Travel Grant, LAUP Department, TAMU, \$600.
2024 – 2028	Graduate Assistantship covering stipend and tuition, Center for Geospatial Sciences, Applications and Technology, TAMU.
2019 – 2022	Academic Scholarship covering all tuition fees, TJU.
2014 – 2019	First-class Scholarships in academic performance (5%, 8 times), SDUST.
2014 – 2019	First-class Corporate Scholarships (5%, 2 times), SDUST.
2014 – 2019	“Outstanding Student Pioneer” (5%, 3 times), SDUST.
2019	“Outstanding Graduate” of Shandong Province (5%).
2019	“Outstanding Student” in Science, Technology, and Innovation (5%), SDUST.
2018	Gold Award, SINO-JAPAN Art & Design Exchange Exhibition.
2018	National Best Leader Award, National Green Building Design Competition.
2018	Grand Prize, National Green Building Design Competition - Deep Green Category.
2018	First Prize, The Fourth Academy Award National Art and Design Exhibition.
2018	Gold Award, Chinese Creative Design Annual 2016/2017.
2018	Silver Award, The Third Bauhaus Award International Design Competition.
2018	Bronze Award, Exhibition of Architectural Design in Developing Countries.
2018	Excellent Award, Shandong Province College Student Construction Design Competition.
2017	Excellence Award, China Architectural Education/TSINGRUN Award Students’ Paper Competition.
2017	Bronze Award, The 3rd Mainland, Taiwan and Hong Kong Design Invitation Exhibition.
2017	Second Prize, National Green Building Design Competition - Green Category.
2016	Excellent Award, National Green Building Knowledge Competition for Youth Students.
2016	Excellent Market Potential Award, Lion Award International Industrial Design Competition –

“Innovative Application Design Group: Industrial Robot Systems”.

TEACHING EXPERIENCE

Teaching Assistant

- 2026 Spring** PLAN626:600: Advanced GIS & PLAN625:600: Introduction to GIS & PLAN625:700: Introduction to GIS (Online), Texas A&M University, College Station, USA
- 2025 Fall** PLAN626:600: Advanced GIS & PLAN625:700: Introduction to GIS (Online), Texas A&M University, College Station, USA
- 2020 Fall** Design Studio III (Urban Planning), Tongji University, Shanghai, China

SERVICES

Professional Membership

American Association of Geographers (AAG)
Association of Collegiate Schools of Planning (ACSP)
American Planning Association (APA)

Invited Reviewer

- International Journal of Geographical Information Science
- Journal of Planning Education and Research
- Landscape and Urban Planning
- Cities
- Journal of Transport Geography
- Sustainable Cities and Society
- Environmental Impact Assessment Review
- Transactions in GIS
- Urban Climate
- Journal of Transport and Land Use
- Transportation Planning and Technology
- Applied Spatial Analysis and Policy
- Computational Urban Science
- Environmental Monitoring and Assessment
- Physics and Chemistry of the Earth
- Frontiers in Environmental Science
- Frontiers in Psychology
- ...

Editorial Boards

- Managing Editor: Computational Urban Science

Conference Organizations

- Session Co-chair: Symposium on Human Dynamics Research: Extreme Weather, Human Responses, and

Urban Resilience (I & II). *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA.

- Session Chair/Primary Organizer: Symposium on Human Dynamics Research: Urban Environmental Intelligence and Human-Climate Interactions (1 & 2). *2025 Annual Meeting of American Association of Geographers*, March 24-28, 2025, Detroit, Michigan, USA.

MEDIA COVERAGE

- TAMIDS Announces 2025-2026 Cohort of Student Ambassadors. <https://tamids.tamu.edu/2025/07/18/tamids-announces-2025-2026-cohort-of-student-ambassadors/>
- Throwing Shade at Heatwaves: Tracking campus “hot spots” could help students stay cool in the Texas heat. <https://engineering.tamu.edu/news/2024/09/throwing-shade-at-heatwaves.html>

SKILLS

- **Programming:** Python (Data Processing, Visualization, and Analysis), SQL, HTML
- **Machine Learning/Deep Learning:** Agentic AI, Computer Vision, Time series Forecasting, Tree models, Explainable AI
- **Spatial Data Analysis:** Google Earth Engine, ArcGIS Pro, QGIS, Geoda, MGWR 2.2
- **Urban Climate Simulation:** SOLWEIG, Envi-met
- **Design and Planning:** AutoCAD, SketchUp, Rhino, Photoshop, Illustrator, InDesign, Enscape, Lumion